

Session Info

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Session

```
sessionInfo()  
  
#> R version 4.5.2 (2025-10-31)  
#> Platform: x86_64-pc-linux-gnu  
#> Running under: Ubuntu 24.04.3 LTS  
#>  
#> Matrix products: default  
#> BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3  
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p0.3.26.so; LAPACK version 3.12.  
#>  
#> locale:  
#> [1] LC_CTYPE=en_US.UTF-8 LC_NUMERIC=C  
#> [3] LC_TIME=en_US.UTF-8 LC_COLLATE=en_US.UTF-8  
#> [5] LC_MONETARY=en_US.UTF-8 LC_MESSAGES=en_US.UTF-8  
#> [7] LC_PAPER=en_US.UTF-8 LC_NAME=C  
#> [9] LC_ADDRESS=C LC_TELEPHONE=C  
#> [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C  
#>  
#> time zone: Etc/UTC
```

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#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats      graphics  grDevices  utils      datasets  methods   base
#>
#> other attached packages:
#> [1] rProject_0.0.25
#>
#> loaded via a namespace (and not attached):
#> [1] backports_1.5.0  R6_2.6.1          lubridate_1.9.5  xfun_0.56
#> [5] magrittr_2.0.4   glue_1.8.0         stringr_1.6.0    knitr_1.51
#> [9] timechange_0.4.0 generics_0.1.4     lifecycle_1.0.5  xml2_1.5.2
#> [13] cli_3.6.5.9000   bibtex_0.5.2       compiler_4.5.2   plyr_1.8.9
#> [17] highr_0.11       rprojroot_2.1.1    httr_1.4.8       tools_4.5.2
#> [21] evaluate_1.0.5   Rcpp_1.1.1         RefManageR_1.4.0 otel_0.2.0
#> [25] rlang_1.1.7      jsonlite_2.0.0     stringi_1.8.7

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Packages

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unname(installed.packages()[, 1])

#> [1] "fitVARMxID"      "metaDyn"          "abind"
#> [4] "Amelia"          "arm"              "arrow"
#> [7] "ash"             "AsioHeaders"      "askpass"
#> [10] "assertthat"      "backports"        "base64enc"
#> [13] "betaDelta"       "betaMC"           "betaNB"
#> [16] "betaSandwich"    "BH"               "bibtex"

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#> [19] "BiocManager"      "BiocVersion"      "bit"
#> [22] "bit64"            "bitops"           "blob"
#> [25] "bootStateSpace"   "brew"             "brio"
#> [28] "broom"            "bslib"            "cachem"
#> [31] "callr"            "car"              "caracas"
#> [34] "carData"          "cellranger"       "cfr"
#> [37] "checkmate"        "cli"              "clipr"
#> [40] "clock"            "clusterGeneration" "coda"
#> [43] "c0de"             "collections"       "colorspace"
#> [46] "commonmark"       "conflicted"        "corpcor"
#> [49] "covr"             "cowplot"          "cpp11"
#> [52] "crayon"           "credentials"       "crosstalk"
#> [55] "cTMed"            "ctsem"            "curl"
#> [58] "data.table"       "DBI"              "dbplyr"
#> [61] "Deriv"            "desc"             "deSolve"
#> [64] "devtools"         "diagram"          "dials"
#> [67] "DiceDesign"       "diffobj"          "digest"
#> [70] "distributional"   "distro"           "doBy"
#> [73] "docopt"           "downlit"          "dplyr"
#> [76] "DT"              "dtplyr"           "duckdb"
#> [79] "dynr"            "dynUtils"         "ellipse"
#> [82] "ellipsis"         "evaluate"         "expm"
#> [85] "fansi"           "farver"           "fastDummies"
#> [88] "fastmap"          "fclust"           "fda"
#> [91] "fdrtool"          "fds"              "fitDTVARmXID"
#> [94] "fitVARmXID"       "FNN"              "fontawesome"
#> [97] "forcats"          "foreach"          "forecast"

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#> [100] "Formula"          "fracdiff"         "fs"
#> [103] "fst"              "fstcore"          "furrr"
#> [106] "future"           "future.apply"     "gargle"
#> [109] "generics"         "gert"             "ggplot2"
#> [112] "ggrepel"          "gh"               "gitcreds"
#> [115] "glasso"           "glmnet"           "globals"
#> [118] "glue"             "googledrive"      "googlesheets4"
#> [121] "gower"            "GPArotation"      "GPfit"
#> [124] "graphicalVAR"     "gridExtra"        "gsubfn"
#> [127] "gtable"           "gtools"           "hardhat"
#> [130] "haven"            "hdcrcde"          "here"
#> [133] "highr"            "Hmisc"            "hms"
#> [136] "htmlTable"        "htmltools"        "htmlwidgets"
#> [139] "httpgd"           "httpuv"           "httr"
#> [142] "httr2"            "ids"              "ifaTools"
#> [145] "igraph"           "infer"            "ini"
#> [148] "inline"           "ipred"            "isoband"
#> [151] "iterators"        "jomo"             "jpeg"
#> [154] "jquerylib"        "jsonlite"         "jsonvalidate"
#> [157] "kernlab"          "knitr"            "ks"
#> [160] "kutils"           "labeling"         "Lahman"
#> [163] "languageserver"   "later"            "latex2exp"
#> [166] "lava"             "lavaan"           "lazeval"
#> [169] "lhs"              "lifecycle"        "lintr"
#> [172] "lisrelToR"        "listenv"          "litedown"
#> [175] "littler"          "lme4"             "lmtree"
#> [178] "locfit"           "longMI"           "loo"

```

#> [181]	"lubridate"	"magick"	"magrittr"
#> [184]	"markdown"	"MatrixModels"	"matrixStats"
#> [187]	"mclust"	"memoise"	"metaDyn"
#> [190]	"metaSEM"	"metaVAR"	"mi"
#> [193]	"mice"	"microbenchmark"	"mime"
#> [196]	"miniUI"	"minqa"	"mitml"
#> [199]	"mize"	"mlVAR"	"mnormt"
#> [202]	"modeldata"	"modelenv"	"modelr"
#> [205]	"MplusAutomation"	"multicool"	"mvtnorm"
#> [208]	"nloptr"	"numDeriv"	"nycflights13"
#> [211]	"OpenMx"	"openssl"	"openxlsx"
#> [214]	"ordinal"	"otel"	"pak"
#> [217]	"pan"	"pander"	"parallelly"
#> [220]	"parsnip"	"patchwork"	"pbapply"
#> [223]	"pbivnorm"	"pbkrtest"	"pcaPP"
#> [226]	"pdftools"	"pillar"	"pkgbuild"
#> [229]	"pkgconfig"	"pkgdown"	"pkgload"
#> [232]	"plyr"	"png"	"posterior"
#> [235]	"pracma"	"praise"	"prettyunits"
#> [238]	"printr"	"processx"	"prodlim"
#> [241]	"profvis"	"progress"	"progressr"
#> [244]	"promises"	"proto"	"ps"
#> [247]	"psych"	"purrr"	"qgraph"
#> [250]	"qpdf"	"quadprog"	"quantreg"
#> [253]	"quarto"	"QuickJSR"	"R.cache"
#> [256]	"R.methodsS3"	"R.oo"	"R.utils"
#> [259]	"R2jags"	"R2WinBUGS"	"R6"

#> [262]	"ragg"	"rainbow"	"rappdirs"
#> [265]	"rbibutils"	"rcmdcheck"	"RColorBrewer"
#> [268]	"Rcpp"	"RcppArmadillo"	"RcppEigen"
#> [271]	"RcppGSL"	"RcppParallel"	"RcppTOML"
#> [274]	"RCurl"	"Rdpack"	"readr"
#> [277]	"readxl"	"recipes"	"RefManageR"
#> [280]	"reformulas"	"rematch"	"rematch2"
#> [283]	"remotes"	"reprex"	"reshape2"
#> [286]	"reticulate"	"rex"	"rhdf5"
#> [289]	"rhdf5filters"	"Rhdf5lib"	"rhub"
#> [292]	"rjags"	"rlang"	"RMariaDB"
#> [295]	"rmarkdown"	"rockchalk"	"roxygen2"
#> [298]	"rpf"	"RPostgres"	"rProject"
#> [301]	"rprojroot"	"rsample"	"RSQLite"
#> [304]	"rstan"	"rstantools"	"rstudioapi"
#> [307]	"RUnit"	"rversions"	"rvest"
#> [310]	"Ryacas"	"S7"	"sass"
#> [313]	"scales"	"selectr"	"sem"
#> [316]	"semlbci"	"semmcci"	"semPlot"
#> [319]	"sessioninfo"	"sfd"	"shape"
#> [322]	"shiny"	"simStateSpace"	"slider"
#> [325]	"snow"	"snowfall"	"sourcetools"
#> [328]	"SparseM"	"sparsevctrs"	"SQUAREM"
#> [331]	"StanHeaders"	"stringi"	"stringr"
#> [334]	"styler"	"symSEM"	"sys"
#> [337]	"systemfonts"	"tailor"	"tensorA"
#> [340]	"testthat"	"texreg"	"textshaping"

```

#> [343] "tibble"          "tidymodels"      "tidyr"
#> [346] "tidyselect"      "tidyverse"       "timechange"
#> [349] "timeDate"        "tinytex"         "tune"
#> [352] "tzdb"            "ucminf"          "unigd"
#> [355] "urca"            "urlchecker"      "usethis"
#> [358] "utf8"            "uuid"            "V8"
#> [361] "vctrs"           "viridisLite"     "vroom"
#> [364] "waldo"           "warp"            "whisker"
#> [367] "whoami"          "withr"           "workflows"
#> [370] "workflowsets"    "xfun"            "XML"
#> [373] "xml2"            "xmlparsedata"    "xopen"
#> [376] "xtable"          "yaml"            "yardstick"
#> [379] "zip"             "zoo"             "base"
#> [382] "boot"            "class"           "cluster"
#> [385] "codetools"       "compiler"        "datasets"
#> [388] "foreign"         "graphics"        "grDevices"
#> [391] "grid"            "KernSmooth"      "lattice"
#> [394] "MASS"            "Matrix"          "methods"
#> [397] "mgcv"            "nlme"            "nnet"
#> [400] "parallel"        "rpart"           "spatial"
#> [403] "splines"         "stats"           "stats4"
#> [406] "survival"        "tcltk"           "tools"
#> [409] "utils"

```

References

R Core Team. (2025). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. <https://www.R-project.org/>